

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	1 June 2026
Team ID	SWTID-2026-8236
Project Name	Credit Card Approval Prediction System
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Credit Card Approval Prediction

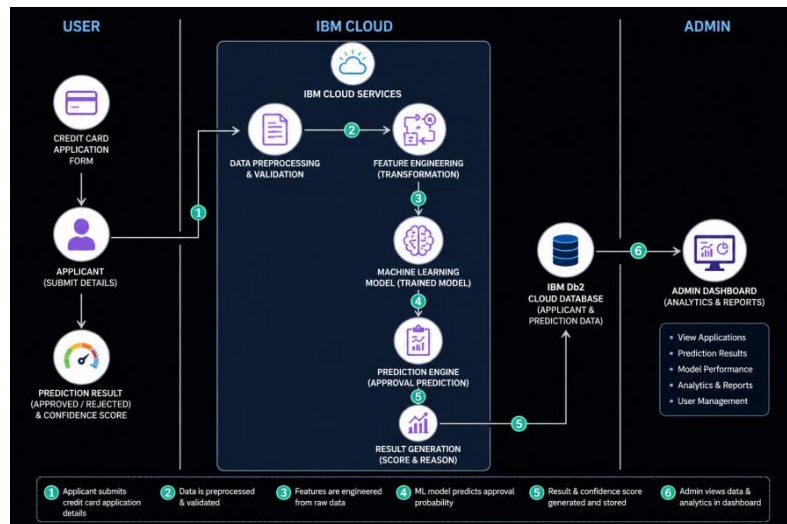


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Applicant form & Admin dashboard	HTML5, CSS3, Bootstrap, JavaScript
2	Application Logic-1	Input validation & preprocessing	Python, Flask
3	Application Logic-2	Feature engineering	Pandas, NumPy
4	Application Logic-3	Credit card approval prediction	Scikit-learn
5	Database	Applicant information	MySQL
6	Cloud Database	Optional deployment	MySQL Cloud / Azure Database
7	File Storage	Dataset & Trained Model (.pkl)	Local Storage
8	External API-1	Email notification (optional)	SMTP
9	External API-2	Credit score API (optional)	Experian/CIBIL API
10	Machine Learning Model	Random Forest (Best Model)	Scikit-learn
11	Infrastructure	Local Server / Cloud	Flask, Windows/Linux, Render/AWS

Table-2: Application Characteristics:

S.No	Characteristic	Technology
1	Open Source Framework	Flask, Scikit-learn, Pandas, NumPy
2	Security	Password Hashing, HTTPS, Input Validation
3	Scalable Architecture	Three-tier Architecture (UI → Flask → Database)
4	Availability	Local Server or Cloud Deployment
5	Performance	Fast prediction (<1 second), Optimized ML model